QP code: **25901641**

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR/IMPROVEMENT/REAPPEARANCE

EXAMINATION APRIL 2026

SECOND SEMESTER

Discipline Specific Core Course (DSC) - MG2DSCCSC100, MG2DSCECC101- PYTHON PROGRAMMING

(2024 ADMISSION ONWARDS)

Duration: 1 Hour 30 Mins **Maximum Marks: 50**

Scheme of Evaluation

Part A

Very Short Answer Questions -Answer all questions - Each question carries 2 marks

1.What is None in Python? [U] / [CO1]
Ans: `None` is a special constant in Python that represents the absence of a value or null value. It is an object of its own datatype `NoneType`.

2. Write a program to find the factorial of a Number [A] / [CO2]
Ans:
```
n = int(input("Enter a number: "))
fact = 1
for i in range(1, n + 1):
  fact = fact * i
print("Factorial =", fact)
```

3 Name the concept in Python that allows a task to be executed multiple times in different parts of a program. [K] / [CO3]

Ans: Functions in Python allow a block of code to be defined once and executed multiple times from different parts of a program. They promote code reusability and modularity.

4. Explain how list concatenation works. [U] / [CO3]
Ans: List concatenation in Python uses the `+` operator to join two or more lists into a new list. It combines elements in order without modifying the original lists.
```
a = [1, 2]
b = [3, 4]
c = a + b    # c becomes [1, 2, 3, 4]
```
Definition -1 mark, Example 1 mark

5. Define set with an example. [K] / [CO3]
Ans: A set in Python is an unordered, mutable collection of unique elements. It does not allow duplicate values and is defined using `{}` or `set()`.
Eg:
```
s = {1, 2, 3, 3, 4}
print(s)
# Output: {1, 2, 3, 4}
```
Definition -1 mark, Example 1 mark

Part B

Short Answer Questions -Answer any 5 questions- Each question carries 5 marks

6. Compare and contrast a Python statement and an expression with examples. [U] / [CO1]

Statement	**Expression**
A statement is an instruction that performs an action.	An expression is a combination of values, variables, and operators that produces a result.
Statements may not return a value.	Expressions always return a value.
Used for controlling program flow, assignments, loops, etc.	Used for calculations and evaluations.
Example: x = 10	Example: x + 5
Statements can contain expressions.	Expressions cannot usually contain statements.

Definition of Statement and Expression – 1 Mark each,
Comparison / Differences – 2 Marks , Examples – 1 Mark

7. Discuss the various features of Python in detail. [K] / [CO1]

Ans: 1. Simple and Easy to Learn: Python has a clean syntax similar to English. It uses indentation instead of braces, making code readable and beginner-friendly.

2. Interpreted Language: Python code is executed line by line by the interpreter. This makes debugging easier and avoids separate compilation.

3. Dynamically Typed: You don't need to declare variable types. The type is determined at runtime.

```
x = 10 # x is int
x = "hello" # now x is str
```

4. High-level Language: Python handles memory management automatically. You don't deal with pointers or memory allocation directly.

5. Portable: Python code runs on any platform like Windows, Linux, Mac without changes, as long as the interpreter is installed. "Write once, run anywhere".

Any 5 features – 1 Mark each

8. Write the difference between break and continue statement in Python. [A] / [CO2]

Ans:

break Statement	**continue Statement**
The break statement is used to terminate the loop immediately.	The continue statement is used to skip the current iteration and continue with the next iteration of the loop.
Control comes out of the loop.	Control remains inside the loop.
Used when the loop needs to stop based on a condition.	Used when a particular iteration needs to be skipped.
Mostly used to exit loops early.	Mostly used to ignore specific values or conditions.

Pogram for break:

```
for i in range(1, 6):
    if i == 3:
        break
    print(i)
```

Output:

1
2

Program for continue:

```
for i in range(1, 6):
    if i == 3:
        continue
    print(i)
```

Output:

1
2
4
5

2.5 marks for each statements

9. Compare the if-else and if-elif-else statements in Python. [U] / [CO2]

Ans:

if-else Statement	**if-elif-else Statement**
Used to check only two conditions.	Used to check multiple conditions.
Contains one if block and one else block.	Contains one if, one or more elif, and an optional else block.
Suitable for simple decision making.	Suitable for multiple decision making.
Execution chooses between two alternatives.	Execution chooses one block among many alternatives.

Syntax of if-else

```
if condition:
    statement1
else:
    statement2
```

Syntax of if-elif-else

```
if condition1:
    statement1
elif condition2:
    statement2
else:
    statement3
```

Difference – 3 marks
Syntax – 2 marks

10. Explain how tuples differ from lists in Python with suitable examples. [U] / [CO3]

Ans:

List	**Tuple**
A list is mutable, which means elements can be changed.	A tuple is immutable, which means elements cannot be changed.
Lists are created using square brackets [ ].	Tuples are created using parentheses ( ).
Lists consume more memory.	Tuples consume less memory.
Lists are suitable for data that may change.	Tuples are suitable for fixed data.

```
# List - mutable

my_list = [1, 2, 3]
my_list[0] = 10 # Works
my_list.append(4) # Works
print(my_list)
```

Output: [10, 2, 3, 4]

```
# Tuple - immutable
my_tuple = (1, 2, 3)
my_tuple[0] = 10 # Error: 'tuple' object does not support item assignment
# my_tuple.append(4) # Error: 'tuple' object has no attribute 'append'
print(my_tuple)
```

Output: (1, 2, 3)

Difference 2.5 marks

Example program -2.5 marks

11. Describe how update() and setdefault() methods work and when to use each, with examples. [U] / [CO3]

Ans: update() method: Updates a dictionary with key:value pairs from another dictionary. If a key exists, its value is overwritten. If it doesn't exist, the new key:value pair is added.
- Used to merge dictionaries or add/update multiple key:value pairs at once.

Example:

```
d = {'a': 1, 'b': 2}
d.update({'b': 5, 'c': 3})
print(d)                    # Output: {'a': 1, 'b': 5, 'c': 3}
```

Here `'b'` was updated and `'c'` was added.

setdefault()method: Returns the value of a key if it exists. If the key doesn't exist, it inserts the key with a specified default value and returns that value.

When you want to get a key's value, but also ensure the key exists in the dictionary with a default if it was missing. Useful for initializing values.

Example:

```
d = {'a': 1, 'b': 2}
val1 = d.setdefault('b', 100) # 'b' exists, so returns 2
val2 = d.setdefault('c', 100) # 'c' missing, adds 'c':100 and returns 100
print(d) # Output: {'a': 1, 'b': 2, 'c': 100}
print(val1, val2)           # Output: 2 100
```

2.5 Marks for each method with example

12. Write a Python program to concatenate two lists [1, 2, 3] and [4, 5, 6] and print the result. [A] / [CO3]

```
list1 = [1, 2, 3]

list2 = [4, 5, 6]

result = list1 + list2

print("Concatenated list:", result)
```

Output

Concatenated list: [1, 2, 3, 4, 5, 6]

5 marks for correct program

Part C

Essay Questions -Answer any 1 questions - Each question carries 15 marks

13. Explain indentation and comments in Python. [U] / [CO1]

Ans: Python uses indentation to define blocks of code. Instead of `{}` like other languages,

Python relies on whitespace at the start of a line. Indentation is mandatory. It tells the interpreter which statements belong to loops, functions, if-else blocks, etc. Standard is 4 spaces per indent level. Mixing tabs and spaces causes `

Comments:

Notes in code that are ignored by the interpreter. Used to explain code for humans.

Types:

1. Single-line comment: Starts with `#`. Everything after `#` on that line is ignored.

2. Multi-line comment: Python has no official syntax, but triple quotes `''' '''` or `""" """` are commonly used. If not assigned, they’re treated as string literals and ignored.

Description on indendation and comments – 6 marks each

Sample Program with proper indendation and Comment – 3 mark

14. Explain the flow of execution in a Python program that includes function definitions and function calls. Describe how control moves between the calling function and the function definition with an example. [U] / [CO3]

Ans: Python is an interpreted language. The interpreter reads code from top to bottom and executes it line by line. However, when functions are present, execution does not always stay linear. Control "jumps" between different parts of the program.

Function definition:When Python sees `def function_name():`, it creates a function object in memory and binds the name to it. The code inside the function is *not executed* at this point.

Function call: When Python sees `function_name()`, it suspends the current flow, jumps to the first line of the function body, executes it, then returns back to the point after the call.

Explanation on function definition and call and flow of control – 8 marks

Program with function definition and call and explanation of control flow in that program -7 marks